



FIRST SEMESTER 2025-26
COURSE HANDOUT Part II

Date: 02.08.2026

In addition to part I (General Handout for all courses appended to the Time table) this portion gives further specific details regarding the course.

Course No : CHE F497

Course Title : Atomic and Molecular Simulations

Instructor-in-Charge : Prof. Sarbani Ghosh (sarbani.ghosh@pilani.bits-pilani.ac.in)

1. Course Description:

This course is to equip students with the knowledge of a variety of modern simulation methods at the atomic and molecular levels. The course will deal with several atomic and molecular simulation methods e.g., ab initio methods, molecular dynamics (MD) simulations, Monte Carlo (MC) methods, free energy algorithms, coarse-graining, and multiscale methods. The course will provide hands-on training for simulations using GROMACS or LAMMPS or similar and hands-on simulations using any DFT packages such as VASP, Quantum Espresso or similar. The course also introduces concepts and algorithms based on programming languages (e.g., Matlab, C, C++, Molecular Modelling Software, and Visual Software). Case studies and examples of several real-life applications will be presented for all the simulation techniques.

2. Scope and Objective of the Course:

This course aims to give an overall introduction to various types of particle-based simulation techniques in the sub-micro level phenomena with hands-on training. Currently, with the advances in computational tools, more and more industries/researchers are focusing on using computational tools to study material properties for various applications. The course will include computational multi-scale methods, from first principle ab initio calculations to molecular mechanics, offering potential pathways for studying advanced materials for several applications such as molecular electronics, energy storage/generation, separation, and purification, to name a few. After successful completion of the course, the students will be equipped with the theoretical basis and algorithms of several atomic and molecular simulations techniques (DFT, MD, MC) with their typical application areas and will be able to solve real-world problems by choosing the right simulation techniques.

3. Course Learning Outcomes:

- Understand and apply fundamental atomic and molecular simulation techniques, including DFT, MD, MC, and multiscale methods.
- Select and apply appropriate simulation approaches to solve real world problems by understanding the phenomena in atomic and molecular scale .
- Analyze and evaluate simulation data to extract meaningful physical and chemical insights for advanced materials applications.

Course Pre/Co- requisite (if any) & Catalogue / Bulletin Description: No prerequisite (Given in the Bulletin)



4. Text Books:

- **(TB1)** Daan Frenkel, Berend Smit. Understanding Molecular Simulation. From algorithm to Applications. 2002. 2nd edition.
- **(TB2)** David Sholl, Janice A Steckel. Density functional theory: A practical introduction. 2009. Wiley
- **(TB3)** M. P. Allen and D. J. Tildesley, Computer simulation of liquids, 1989, Clarendon Press. NY, USA

5. Reference Books:

1. Marx and Hütter, Ab Initio Molecular Dynamics
2. Maria Velinova, Coarse-Grained Molecular Dynamics, 2017
3. Donald A. McQuarrie. Statistical Mechanics. 2000. University Science Books
4. Statistical Mechanics: Theory and Molecular Simulation Mark Tuckerman, Oxford University Press (2010)
5. Molecular Modelling: Principles and Applications (2nd edition) Andrew R. Leach, Prentice-Hall (2001).

6. Course Plan:

Module No.	Lecture Session	Reference	Learning outcomes
1. Introduction to atomic and Molecular Simulation Methods and recap of Statistical Thermodynamics	L1-L5: Overview of Atomic and Molecular Simulation Methods Fundamental of Statistical Mechanics, Quantum Mechanics, Review of Probability, Micro State, Macro State, Thermodynamic Ensembles	Text Book (TB) TB1-Ch1 TB1-Ch2 TB3-Ch2	• Discuss various aspects of atomic and molecular simulations • Recall the statistical and quantum mechanics used in atomic and molecular simulations
2. Quantum Chemical Methods	L6-L11: Born-Oppenheimer Approximation, Hydrogen and Multielectron Atoms, Hartree-Fock Theory and Basis Sets, Density Functional Theory, Geometry Optimization, Electronic Structure of Atoms and Molecules, Ab Initio MD, Application, A Case Study	TB2-Ch1, Ch3 RB-1	• Discuss the theoretical basis of abinitio calculations • Discuss the significance of basic algorithms and their parameters for a successful simulation
3. Introduction to Quantum Espresso	L12-L14: Hands on simulations using DFT package, e.g., Gaussian, Quantum Espresso	Class notes	• Independently design, run and analyse DFT simulations in a relatively simple scientific problem
4. Introduction to Molecular Editing and Visualization Software	L15-L17: Making Model Structure, Viewing Simulation Structures and Movies, Making Movies from	Class notes	• Identify a system and create the model structure



	Simulation Trajectories		• visualize the structures and MD/MC trajectories
5. Molecular Dynamics (MD) Simulations	L18-L22: Basics of Molecular Dynamics, Integrators for Newton's Equations, Time step and Timescale Considerations, Thermostat, Barostat, Constant Temperature and Pressure MD, Practical Aspects of Simulations: Simulation of Bulk Phases, Periodic Boundaries, Initialization, Force Computation, Long-Range Interactions, Potential Energy Truncation, Minimum Image Convention	TB1-Ch4 TB3-Ch3	• Discuss the theoretical basis of MD • Discuss the method well enough to be able to deepen the knowledge of various aspects of MD
6. Model and Force Field	L23-L25 : Classical Force Field, Bonded and Nonbonded Interactions, Implicit Solvation, Output and Analysis, Application, A Case Study	TB1-Ch3 TB3-Ch1	• Discuss the significance of model, force field and other parameters for a successful simulation
7. MD simulation using LAMMPS/GROMACS	L26-L33: Hands on simulations using MD simulation package, e.g., LAMMPS, GROMACS	Reference book 3 + class notes	• Independently design, run and analyse simulations in a relatively simple scientific problem • Learn independent project work and solving open real-world problems
8. Monte Carlo (MC) Methods	L34-L37: Introduction to Monte Carlo Simulation, Markov Chains, The Metropolis Method, Random Number Generator, MC Simulation in Various Ensembles: Canonical Ensemble MC, Isobaric-Isothermal Ensemble, Grand Canonical Ensemble, Rigid Molecules and Orientation Moves, Equilibration, Hybrid MC/MD, Application, A Case Study	TB1-Ch3 TB3-Ch4	• Discuss the theoretical basis of MC • Discuss the method well enough to be able to deepen the knowledge of MC and to run simulations independently
9. Introducing Coarse Graining and Multiscale Techniques	L37-L40: Kinds of Simplified Models, Advantages of Coarse Graining, Development and	TB3 + Class notes	• Explain the hurdles of simulations in atomic scale



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	Parameterization of Coarse-Grained Models		• Describe coarse-grained models • Elaborate the basics of coarse-grain parameterization
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7. Evaluation Scheme:

Component	Duration	Weightage (%)	Date & Time	Nature of component (Close Book/ Open Book)
Surprised Class Tests (Best 3 out of 4)	20 min	10	In Class	CB
1 Announced Class Test (Pre-Midsem)	25 min	5	In Class TBA	CB
Assignment*-I (Based on Pre-MidSem contents)	1 week	15	TBA	Take Home
Assignments*-II (1 or 2)	2 week	30	TBA	Take Home
Comprehensive Examination	180 min	40	TBA	CB and/or OB

*Assignments: This component involves home assignments throughout the semester based on hands-on training of several simulation techniques.

8. Chamber Consultation Hour: Will be announced in the class.

9. Notices: All Notices would be put up on Google classroom/Nalanda and/or be announced during the class. All students attending this course **MUST** join google classroom with the class ID TBA . All course related notices and announcements will be announced in the class or be posted in the classroom/Nalanda. Assignments may also be posted in classroom/Nalanda.

10. Make-up Policy: Make-up is granted only for genuine cases in the case of hospitalizations or when complete bed-rest is medically advised with proper justifications, valid documents, and prior permission from the Instructor-in-charge, unless its an emergency. No make-up for class tests and assignments.

11. Note (if any):

*Use of **computer** is mandatory to learn simulations.*

Sarbani Ghosh

Instructor-in-charge

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